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Comparative Evaluation of Deep Sequence-Learning Models for Multi-Output Electricity Demand Forecasting in Sustainable Power Systems

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Abstract: Accurate electricity demand forecasting is important for power system planning, grid operation, and the integration of variable renewable energy sources. Although deep sequence-learning models are widely used for load forecasting, the relative performance of recurrent, gated, and hybrid architectures remains dependent on the characteristics of the available data and the forecasting procedure. This study investigates the performance of eight deep sequence-learning architectures for the simultaneous prediction of three electricity demand series in Great Britain. Daily demand records from 1 January 2018 to 30 November 2023 were used to develop and evaluate recurrent neural network (RNN), bidirectional RNN, long short-term memory (LSTM), bidirectional LSTM, gated recurrent unit (GRU), bidirectional GRU, convolutional neural network (CNN)-LSTM, and CNN-GRU models. All models used a 12-day input window and were evaluated using a chronological 80/20 training–test split. A separate 366-day recursive forecasting exercise was also conducted for the period from 1 December 2023 to 30 November 2024. The conventional RNN achieved the lowest mean absolute error (MAE) across National Demand, Transmission System Demand, and England and Wales Demand, with an average of approximately 1,246 MW. Its corresponding errors were 1,151 MW, 1,517 MW, and 1,069 MW, respectively. More complex gated, bidirectional, and convolutional-recurrent architectures produced higher errors under the implemented training and evaluation procedures. These findings indicate that greater architectural complexity does not necessarily translate into more accurate electricity demand predictions for daily-aggregated load data. The comparison also identifies the importance of model configuration and output handling when interpreting differences in forecasting performance. The study provides empirical evidence to inform model selection for electricity demand forecasting, an essential component of data-driven planning and operational decision-making in sustainable power systems.

Keywords: Electricity demand forecasting; Sustainable power systems; Deep sequence learning; Recurrent neural networks; Multi-output forecasting; Smart grids

1 Introduction

Electric power systems require a continuous balance between electricity generation and consumption. Although energy storage technologies are increasingly deployed, their capacity, cost, and operating characteristics still limit the extent to which they can compensate for variations in supply and demand across an entire power system. Reliable electricity demand forecasting therefore remains an important requirement for generation scheduling, network planning, system balancing, and the operation of electricity markets [1]. Forecasting requirements differ considerably across operational timescales, from intraday adjustments to longer-term planning decisions. Statistical and econometric methods have traditionally provided the basis for these forecasts, but the increasing availability of historical electricity data has encouraged the use of machine-learning approaches capable of representing nonlinear relationships and changing demand patterns [2, 3]. Recurrent neural networks (RNNs) and related sequence-learning architectures are particularly relevant because electricity consumption exhibits temporal dependence at several scales. Nevertheless, greater model complexity does not necessarily produce more accurate forecasts, especially when the available observations are limited or the temporal resolution does not capture the variability that more elaborate architectures are designed to represent.

Great Britain provides a useful setting for examining these questions. Historical records published by the National Grid Electricity System Operator include half-hourly measurements of National Demand (ND), Transmission System Demand (TSD), and England and Wales Demand (EWD), together with information on embedded wind and solar generation [4]. These demand measures describe related but distinct aspects of electricity consumption and transmission-system requirements, making them suitable for examining whether a common forecasting model can represent several correlated targets. Their relevance has also increased as the British electricity system has moved towards a generation mix with a larger contribution from variable renewable sources. Wind and solar generation introduce additional uncertainty into the balance between electricity supply and consumption, while periods of low demand and high renewable output can create difficulties for conventional generation scheduling and market operation. Negative wholesale electricity prices provide one indication of these changing operating conditions; 107 one-hour periods of negative prices were reported in Great Britain during 2023 [5]. Related challenges are illustrated by the so-called duck curve, in which substantial daytime solar generation reduces net demand before a relatively steep increase in conventional generation requirements later in the day [6, 7]. Electricity demand forecasts cannot, by themselves, resolve renewable generation variability or prevent curtailment. They nevertheless provide information needed for coordinating generation resources, assessing future system requirements, and planning the operation of flexible electricity demand alongside renewable supply [8].

The importance of demand forecasting extends beyond conventional generation and transmission planning. Industrial electricity consumers, including precision-manufacturing and digital-fabrication facilities, contribute to system demand and may provide opportunities for greater flexibility in electricity consumption. Computer numerical control machining centres, additive-manufacturing equipment, and robotic production systems often operate under tightly specified process conditions, and interruptions or disturbances in electricity supply can affect production continuity and product quality [9, 10]. At the same time, some industrial operations can be scheduled within defined production constraints, allowing electricity consumption to be shifted between operating periods. Previous research on energy-aware manufacturing has examined how machine-level electricity requirements, production schedules, and operating costs can be considered jointly, including the scheduling of additive-manufacturing processes under energy-related constraints [10, 11]. Such applications illustrate the wider relevance of electricity demand information to demand-side management. However, national electricity demand forecasts and facility-level consumption forecasts address different operational problems and cannot be treated as interchangeable. The present study concentrates on grid-level demand prediction rather than manufacturing-process optimisation. Its relevance to industrial energy management lies in the general forecasting problem and the comparison of modelling approaches that may subsequently be examined using facility-level electricity data. More broadly, reliable forecasts of correlated demand variables can contribute to the information available for energy management, flexible-load planning, and operational decisions in power systems with increasing renewable generation.

Against this background, this study investigates the comparative forecasting performance of eight deep sequence-learning architectures using three electricity demand series from Great Britain. The analysis covers daily-aggregated observations from 1 January 2018 to 30 November 2023 and considers ND, TSD, and EWD within a shared-input, multi-output forecasting framework. Rather than constructing an independent model for each demand measure, the multi-output formulation allows the three related series to be represented through common network parameters, while retaining separate predictions for each target [12]. The architectures examined comprise a conventional RNN, a bidirectional RNN (Bi-RNN), a long short-term memory (LSTM) network, a bidirectional LSTM (Bi-LSTM), a gated recurrent unit (GRU) network, a bidirectional GRU (Bi-GRU), and two convolutional-recurrent configurations, convolutional neural network (CNN)-LSTM and CNN-GRU. The hybrid models combine one-dimensional convolutional feature extraction with stacked recurrent layers. All eight models use a 12-day historical input window, training-based Min–Max scaling, and the Huber loss function. Their forecasting performance is assessed on a chronologically separated test partition using mean squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE), with mean absolute percentage error (MAPE) also defined in the evaluation framework. A separate recursive forecasting exercise extends the model predictions over 366 days, from 1 December 2023 to 30 November 2024. The comparison is intended to establish how the selected architectures perform under the implemented data-processing and training procedures, rather than to assume that deeper or more elaborate networks necessarily provide superior predictions. Differences in model configuration and output handling are also considered when interpreting the observed performance.

Recent comparative evidence indicates that the relative accuracy of recurrent, convolutional, and gradient-boosting approaches varies across electricity demand datasets, with no single architecture consistently outperforming the alternatives under all conditions [13]. This finding raises a practical question for electricity demand forecasting: whether additional architectural complexity is justified when several strongly related demand series must be predicted from a relatively limited daily dataset. The present study addresses this question through a common multi-output forecasting task, rather than by introducing a new neural network architecture. Its contribution lies in the direct comparison of eight sequence-learning configurations across three related electricity demand measures,

the examination of their target-specific prediction errors, and the assessment of how model configuration and evaluation procedures affect the interpretation of forecasting results. The analysis also distinguishes one-step test-set performance from the behaviour of recursive predictions over a substantially longer horizon. These distinctions are relevant when selecting forecasting models for data-driven electricity system planning, particularly where model complexity must be considered alongside predictive accuracy and the characteristics of the available data. The remainder of the paper is organised as follows. Section 2 reviews statistical, machine-learning, and deep-learning approaches to electricity demand forecasting, together with related research on energy-aware manufacturing. Section 3 describes the dataset, preprocessing procedures, forecasting architectures, and evaluation methods. Section 4 presents the experimental results, including the test-set comparisons and recursive forecasts. Section 5 discusses the findings and their implications for sustainable power systems, while Section 6 summarises the conclusions and directions for further research.

2 Literature Review

2.1 Statistical and Classical Machine-Learning Approaches

Statistical time-series models have long been used for electricity demand forecasting, particularly the autoregressive integrated moving average (ARIMA) family and its seasonal extensions. These methods remain relevant because their underlying assumptions are relatively transparent and their parameters can be interpreted in relation to the temporal structure of the data. They can also provide competitive forecasts when demand patterns are sufficiently regular and the principal dependencies can be represented through linear relationships [14–16]. However, the usefulness of a forecasting model depends not only on its average prediction error but also on the decisions for which its predictions are intended. Hong and Fan [1] examined probabilistic load forecasting and argued that point predictions alone cannot adequately represent the uncertainty faced by electricity system operators. Their review established the importance of forecasting uncertainty when predicted demand is used for operational planning and decision-making. In a broader comparison of conventional and artificial-intelligence-based forecasting methods, Wei et al. [17] reported that statistical approaches, including ARIMA and regression models, remained competitive for certain forecasting horizons, whereas artificial neural networks (ANNs), random forests (RFs), and support vector regression (SVR) were better suited to some problems involving nonlinear demand relationships. Similar qualifications were identified by Yildiz et al. [18], who examined regression and machine-learning methods for electricity demand prediction in commercial buildings. Their findings showed that appropriately specified linear regression models could perform as well as, or better than, more elaborate alternatives under particular data conditions. Taken together, these studies suggest that forecasting accuracy cannot be inferred from model complexity alone; temporal resolution, predictor availability, and the characteristics of the demand series must also be considered.

Classical machine-learning methods have provided further alternatives to statistical forecasting. Support vector machines (SVMs), RFs, gradient boosting machines (GBMs), and extreme gradient boosting (XGBoost) can represent nonlinear relationships without requiring a predetermined linear form [17, 19]. Their application to electricity demand forecasting commonly involves the construction of lagged variables, calendar indicators, weather-related predictors, or other explanatory features. The quality of these inputs can substantially influence prediction accuracy, while the selection of hyperparameters introduces additional modelling decisions. Although such methods are capable of representing complex demand relationships, their performance remains dependent on the information available for training and the conditions under which predictions are evaluated [17, 19]. This dependence is particularly relevant when forecasting is based primarily on historical demand observations rather than a broader set of external variables.

2.2 Recurrent and Gated Neural Architectures

ANNs were introduced into electricity load forecasting several decades ago, with early studies demonstrating their ability to represent nonlinear relationships between historical observations and future electricity demand [20]. Subsequent applications extended neural-network forecasting to residential and commercial energy consumption, where demand patterns reflect a combination of temporal behaviour, building characteristics, and operating conditions [21]. RNNs provide a different approach to sequence modelling by maintaining a hidden state that is updated as successive observations are processed. This structure allows information from preceding time steps to influence later predictions and is therefore applicable to electricity demand series in which recent consumption patterns contain information about subsequent values [22]. Conventional RNNs nevertheless encounter difficulties when learning dependencies over long sequences. During gradient-based training, information propagated through repeated recurrent operations may diminish or grow excessively, making long-range relationships difficult to estimate reliably [23]. The LSTM architecture introduced by Hochreiter and Schmidhuber [24] addressed this limitation through a memory cell controlled by input, forget, and output gates. By regulating the retention and updating of information, LSTM networks can represent temporal dependencies that are difficult for conventional recurrent structures to learn. Their application to electricity demand forecasting has consequently been examined across different consumption scales and forecasting horizons [25–27].

The empirical performance of LSTM models has been investigated using datasets that differ considerably in size, frequency, and geographical coverage. Berriel et al. [28] examined monthly electricity consumption forecasting using more than 21 million consumption records from Brazil. Their implementation combined sliding-window processing with GPU-supported training, and the reported results favoured LSTM over the classical regression methods considered, particularly for longer forecasting horizons. Alanbar et al. [29] compared LSTM with ANN, SVM, and adaptive neuro-fuzzy inference system (ANFIS) approaches for daily energy consumption forecasting in Saudi Arabia. Their study reported prediction accuracy exceeding 94% under the evaluation conditions used, including experiments with limited training data. Rodrigues et al. [30] investigated electricity consumption forecasting using approximately eighteen months of hourly observations from 93 households in Lisbon. Their results indicated that neural-network methods applied to relatively simple metering arrangements could achieve prediction accuracy comparable to approaches requiring more extensive measurement information. Although these studies support the application of neural networks to energy demand prediction, their results also reflect differences in data aggregation, forecasting horizon, and evaluation criteria. Such differences limit the extent to which performance rankings can be transferred directly between applications.

The GRU, introduced by Cho et al. [31], provides an alternative recurrent structure with fewer gating operations than the LSTM. It combines information retention and state updating through reset and update gates without maintaining a separate memory cell. Chung et al. [32] subsequently examined gated recurrent architectures and reported that GRU models could achieve performance comparable to LSTM networks on several sequence-learning tasks, while requiring fewer parameters under comparable configurations. These characteristics have supported the use of GRU-based approaches in electricity load forecasting, including applications where computational requirements are an important consideration [33]. Bidirectional recurrent networks offer another architectural variation. Originally developed by Schuster and Paliwal [34], these networks process an input sequence in both temporal directions and combine the resulting hidden representations. In forecasting applications, bidirectional processing can extract information from different positions within an observed historical window without requiring observations beyond the prediction origin. Related recurrent pooling approaches have been examined for electricity load forecasting, where the representation of temporal information within the input sequence can influence predictive performance [35]. However, the additional parameters and processing operations associated with gated or bidirectional architectures do not establish that they will outperform simpler recurrent models on every electricity demand dataset.

2.3 Convolutional and Hybrid Architectures

CNNs have also been applied to electricity demand forecasting, particularly through one-dimensional convolutional operations that extract local patterns from sequential observations. Unlike recurrent networks, convolutional models identify temporal features through filters applied across neighbouring input positions. Parameter sharing allows these filters to detect similar patterns at different locations within a sequence while limiting the number of independently estimated weights [36]. Their performance has been examined in household and electricity system forecasting applications. Acharya et al. [37] investigated CNN-based short-term load forecasting for an individual household and reported better performance than the LSTM configurations considered after introducing data augmentation. Gasparin et al. [38], however, found that recurrent models achieved stronger short-term forecasting accuracy than CNN-based alternatives in their electricity load experiments, although convolutional approaches offered advantages in inference speed. These contrasting results indicate that the relative performance of convolutional and recurrent structures depends on the forecasting task and the way temporal information is represented.

Hybrid convolutional-recurrent models attempt to combine local temporal feature extraction with recurrent sequence modelling. In such architectures, convolutional layers generally process the input observations before the resulting feature representations are passed to LSTM or GRU layers. Tian et al. [39] developed a CNN-LSTM model for short-term electricity load forecasting and reported error reductions of approximately 9–14% relative to the standalone CNN and LSTM configurations examined in their study. Farsi et al. [40] proposed parallel LSTM-CNN network, a parallel LSTM-CNN architecture evaluated using electricity load datasets from Malaysia and Germany. Their results demonstrated the applicability of combining recurrent and convolutional components, although the additional architectural complexity did not lead to proportional improvements under every experimental condition. Beyond conventional convolutional-recurrent combinations, Fan et al. [41] investigated multivariate time-series prediction using parallel spatio-temporal attention and stacked temporal convolutional networks (TCNs). Their approach produced lower forecasting errors than the ARIMA, SVM, and multilayer perceptron baselines included in the comparison. Lara-Benítez et al. [42] examined TCN and LSTM models using electricity grid and electric-vehicle charging datasets from Spain. Their findings indicated that temporal convolutional models could provide faster inference and favourable performance over longer forecasting horizons. In contrast, Lemos et al. [43] reported that hyperparameter optimisation of a TCN produced only a modest improvement of approximately 1% over the corresponding unoptimised configuration in their electricity consumption forecasting experiments. Gradient boosting and ARIMA also performed better under some of the evaluated conditions. The differences among these

studies suggest that convolutional feature extraction and deeper network structures should be assessed in relation to the available data rather than treated as inherently more accurate forecasting solutions.

A more directly comparable investigation was reported by Musbah and Elsaraiti [13], who evaluated RNN, LSTM, GRU, TCN, category boosting, XGBoost, and LightGBM models using six hourly electricity load datasets from different geographical settings. Their findings showed considerable variation in forecasting performance across regions and dataset durations. Conventional RNN models performed competitively among the deep-learning approaches, while gradient-boosting methods generally provided more consistent results across the datasets considered. This comparison is relevant to the present study because it questions the assumption that gated or more elaborate sequence-learning architectures necessarily provide the lowest forecasting errors. However, the earlier investigation focused on hourly load forecasting across separate datasets, whereas the present analysis considers three related demand measures within a common multi-output framework using daily-aggregated observations. The distinction is important because temporal aggregation, target dependence, and model output configuration may affect the relative performance of recurrent and hybrid forecasting architectures.

2.4 Energy Demand Forecasting and Demand-Side Management in Sustainable Power Systems

Electricity demand forecasting is also relevant to the management of flexible consumption in sustainable power systems. As the contribution of variable renewable generation increases, the ability to anticipate demand becomes important for coordinating electricity supply, identifying periods of system stress, and evaluating opportunities to shift consumption within operational constraints. Industrial facilities provide one possible setting for such flexibility because certain production activities can be rescheduled without necessarily changing the required production output. Prabhu et al. [10] examined energy-aware manufacturing operations and identified machine-level electricity consumption and energy-related operating conditions as factors that should be considered alongside conventional production objectives. Their work established a connection between production planning and electricity use, particularly where energy requirements vary across machines, operations, and scheduling periods. Karimi et al. [11] subsequently investigated energy-aware production scheduling for additive manufacturing. Their study considered the relationship between process settings, job sequencing, electricity demand, and production constraints, showing that energy-related objectives could be incorporated into manufacturing scheduling decisions. These studies illustrate how electricity consumption information can be used within broader operational optimisation problems, although their principal focus is facility-level production rather than national electricity demand forecasting.

The relationship between forecasting and energy management has become more prominent with the development of digitally connected industrial systems. A review of advanced smart manufacturing applications [9] discussed the use of digital twins, cyber-physical systems, and predictive analytics in production monitoring and operational planning. Such technologies provide a basis for linking electricity consumption information with decisions concerning equipment operation, production timing, and resource allocation. In principle, forecasts may assist industrial consumers in identifying periods suitable for flexible-load operation or participation in demand-response arrangements. They may also inform scheduling decisions when electricity prices or renewable generation availability vary over time. However, these applications require information beyond electricity demand forecasts alone, including facility-level consumption characteristics, production constraints, electricity prices, and, where relevant, renewable generation forecasts. Consequently, evidence obtained from ND prediction cannot establish the effectiveness of a particular manufacturing scheduling strategy without further application-specific testing.

The present study therefore addresses a forecasting problem that precedes, but is distinct from, operational energy management. Its primary concern is the relative performance of sequence-learning architectures when several correlated electricity demand variables are predicted simultaneously. The literature reviewed above establishes the relevance of statistical, recurrent, gated, and convolutional approaches, but also shows that reported performance depends on data characteristics, model configuration, and evaluation design. The comparison undertaken here examines these issues using daily electricity demand observations from Great Britain and a shared multi-output prediction task. Its results are intended to inform the selection and interpretation of electricity demand forecasting models within sustainable power system analysis, while providing a methodological reference for future studies that connect demand prediction with renewable energy integration, flexible consumption, or industrial energy management.

3 Methodology

3.1 Data Source and Preprocessing

The electricity demand records were obtained from the National Grid Electricity System Operator for Great Britain. The dataset version used in this study contains observations up to 30 November 2023 and was accessed via Kaggle [4]. The original dataset contains half-hourly observations of electricity demand, embedded renewable generation, pumped-storage operation, interconnector flows, and related system variables. Three demand measures were selected as forecasting targets: ND, TSD, and EWD. ND represents demand derived from metered generation

after excluding specified station loads, pumped-storage pumping, and interconnector exports. TSD includes additional system requirements beyond ND, while EWD represents demand within England and Wales.

The analysis covered 1 January 2018 to 30 November 2023. Half-hourly observations were averaged by calendar date, yielding 2,160 daily observations for each target. This period included seasonal demand variation and the disruption associated with the coronavirus disease 2019 pandemic. Daily aggregation reduced intraday fluctuations and provided a consistent temporal resolution for comparing the forecasting models. Figure 1 presents the three demand series, and Table 1 reports their descriptive statistics. Mean demand ranged from 25,536.0 MW for EWD to 29,562.6 MW for TSD, with coefficients of variation of approximately 16% across the three series.

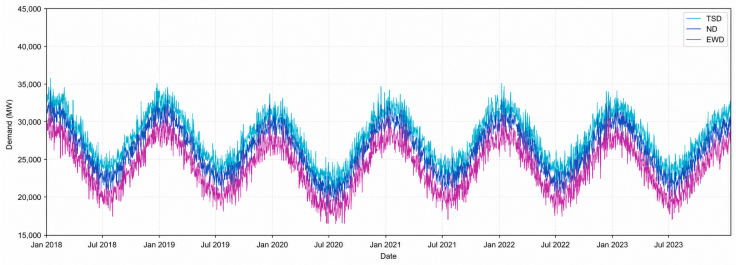


Figure 1. Daily-aggregated ND, TSD, and EWD in Great Britain, 1 January 2018–30 November 2023

Note: ND = National Demand; TSD = Transmission System Demand; EWD = England and Wales Demand.

Table 1. Descriptive statistics of daily electricity demand series (2018–2023)

Statistic	ND	TSD	EWD
Count	2,160	2,160	2,160
Mean (MW)	27,940.4	29,562.6	25,536.0
Standard deviation (MW)	4,477.2	4,396.4	4,118.8
Minimum (MW)	17,560.2	19,465.7	16,450.6
25th percentile (MW)	24,508.3	26,365.1	22,362.1
Median (MW)	27,523.1	29,070.5	25,094.4
75th percentile (MW)	31,172.1	32,605.2	28,611.8
Maximum (MW)	40,297.2	42,152.3	37,573.8

Note: ND = National Demand; TSD = Transmission System Demand; EWD = England and Wales Demand.

The dataset was divided chronologically into training and test partitions at a ratio of 80:20, without random shuffling. Min–Max scaling was performed separately for each target using parameters calculated exclusively from the training partition. The same parameters were applied to the test data to prevent information leakage. Normalisation was performed as follows:

$$x' = \frac{x - x_{\text{min}}^{\text{train}}}{x_{\text{max}}^{\text{train}} - x_{\text{min}}^{\text{train}}} \tag{1}$$

where, x and x' denote the original and normalised observations, respectively, and $x_{\text{min}}^{\text{train}}$ and $x_{\text{max}}^{\text{train}}$ are the minimum and maximum values in the training partition.

A sliding window containing 12 consecutive daily observations was used to predict demand on the following day. The three normalised series were combined into an input tensor of shape (N, 12, 3), where, N is the number of samples. Each target vector contained the subsequent-day values of ND, TSD, and EWD. For the initial test predictions, the preceding training observations were retained as historical inputs, without introducing information from future test observations.

3.2 Multi-Output Forecasting Framework

A shared-input, multi-output framework was adopted to predict ND, TSD, and EWD simultaneously. These demand measures are closely related because they describe overlapping components of the same electricity system. Their temporal dependence provides a basis for learning common representations rather than constructing three entirely independent forecasting networks [12].

Each model received the same 12-day historical window and generated three next-day demand predictions. Shared representations may capture patterns common to the three targets, although differences in their temporal behaviour

can also introduce interference between prediction tasks [44]. The multi-output formulation was therefore treated as a common experimental framework rather than an independently established improvement over single-output forecasting.

3.3 Deep Sequence-Learning Architectures

Eight sequence-learning architectures were implemented in TensorFlow/Keras: conventional RNN, Bi-RNN, LSTM, bidirectional LSTM (Bi-LSTM), GRU, Bi-GRU, CNN-LSTM, and CNN-GRU.

The conventional RNN used recurrent hidden-state updates to represent temporal dependencies in the input sequence [22]. Bi-RNN processing combined forward and backward representations of the observed historical window [34]. LSTM networks employed memory cells and input, forget, and output gates to regulate information retention [24] and address limitations associated with learning long-term dependencies [23]. Bi-LSTM extended this structure through bidirectional processing [35].

GRU networks used update and reset gates without maintaining a separate memory cell [31, 32]. The Bi-GRU architecture applied the same gating mechanism in both temporal directions [31]. The CNN-LSTM and CNN-GRU models combined three one-dimensional convolutional layers with stacked recurrent layers. The convolutional stages extracted local temporal patterns before recurrent processing [39, 40], with increasing filter depth supporting hierarchical feature representation [36].

All models used the same input shape, although their network depths, learning rates, training durations, and output-processing procedures differed. Table 2 summarises the implemented configurations.

Table 2. Architecture configurations and training parameters

Model	Network Configuration	Learning Rate	Epochs
RNN	SimpleRNN(200, seq) → SimpleRNN(200) → Dense(125) → Dense(3)	10^{-6}	300
Bi-RNN	Bi-RNN(300→200→100→45) → Dense(45→10→3)	10^{-7}	200
LSTM	LSTM(80) → LSTM(60) ×4 → Dense(184→162→127→56→36→27→9→6→3)	10^{-4}	300
Bi-LSTM	Bi-LSTM(100) ×3 → Bi-LSTM(60) ×2 →Dense(184→162→127→56→36→27→9→6→3)	10^{-4}	300
GRU	GRU(100) ×2 → Dense(50→25→9→3)	10^{-4}	300
Bi-GRU	Bi-GRU(100) ×3 → Dense(50→25→9→3)	10^{-3}	100
CNN-LSTM	Conv1D(32→64→128) → MaxPool(2) → Bi-LSTM(120) ×3 → Bi-LSTM(100) → LSTM(100) → Dense(84→77→44→22→9→3)	10^{-5}	60
CNN-GRU	Conv1D(32→64→128) → MaxPool(2) → Bi-GRU(120) ×3 → Bi-GRU(100) → LSTM(100) → Dense(84→77→44→22→9→3)	10^{-4}	100

Note: seq indicates that the temporal dimension is retained. All convolutional layers used a kernel size of 3 and causal padding. Repeated layers are indicated by their multiplicity. The sequence-output settings and model-specific prediction procedures are described in Section 3.6; RNN = recurrent neural network; Bi- = bidirectional; LSTM = long short-term memory; GRU = gated recurrent unit; CNN = convolutional neural network.

3.4 Model Training

All models were trained using the Adam optimiser with $\epsilon = 10^{-4}$ and the Huber loss function. A batch size of one was used, and training samples were processed chronologically with shuffle=False. The Huber loss was selected because its quadratic treatment of small residuals and linear treatment of larger residuals limits the influence of unusually large forecasting errors.

Learning rates were selected separately for each architecture using an exponential learning-rate range test [12]. The learning rate was increased from 10^{-9} over approximately 50–100 epochs, and a value below the region of loss divergence was selected for the main training run. Maximum training durations ranged from 60 to 300 epochs, as reported in Table 2.

Early stopping monitored training MAE with a patience of seven epochs and became active after epoch 70. Model checkpoints retained the weights associated with the lowest monitored training error. Because no independent validation criterion was reported for this procedure, the resulting configurations should not be interpreted as having undergone validation-based hyperparameter optimisation.

3.5 Forecast Evaluation

Model predictions were inverse-transformed into MW using the scaling parameters fitted on the training data. Forecasting accuracy was assessed separately for ND, TSD, and EWD using MSE, RMSE, MAE, and MAPE.

The evaluation metrics were defined as follows:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{3}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{4}$$

$$\text{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{5}$$

where, n is the number of test observations, y_i is the observed demand, and $\hat{y}_i$ is the corresponding prediction. MSE is expressed in MW^2, RMSE and MAE in MW, and MAPE as a percentage. The quantitative comparison presented in Section 4 is based on MSE, RMSE, and MAE.

Following the chronological test evaluation, the trained models were used to generate a 366-day recursive forecast from 1 December 2023 to 30 November 2024. Starting from the final available 12-day input window, each predicted demand vector was incorporated into the subsequent input, replacing the oldest observation. This process continued until predictions had been generated for the full period. The exercise examined the behaviour of the models over an extended prediction horizon but did not constitute an independently validated long-term accuracy assessment.

3.6 Output Handling and Evaluation Consistency

The GRU and Bi-GRU implementations retained the temporal dimension in their final recurrent outputs, producing tensors of shape (N, 12, 3). These outputs were averaged across the 12 time steps to obtain one three-variable prediction per sample. The remaining six architectures generated predictions of shape (N, 3) directly and did not require this additional operation.

Although the resulting prediction vectors had a common format, the output-processing procedures were not identical. Temporal averaging may have influenced the errors reported for GRU and Bi-GRU. Accordingly, the experiment compared eight implemented forecasting configurations rather than isolating the effects of recurrent architecture alone. A fully controlled comparison would require consistent output construction and comparable hyperparameter-selection procedures.

4 Results

4.1 Test-Set Forecasting Performance

The forecasting performance of the eight sequence-learning architectures was evaluated using the chronological test partition, which comprised approximately the final 20% of the daily electricity demand observations from 2018 to 2023. Table 3 presents the MSE, RMSE, and MAE for ND, TSD, and EWD.

The conventional RNN produced the lowest error across all three demand series. For ND, its MSE, RMSE, and MAE were 2,145,599 MW^2, 1,465 MW, and 1,151 MW, respectively. The corresponding MAE values for TSD and EWD were 1,517 MW and 1,069 MW. Averaged across the three targets, the RNN achieved an MAE of approximately 1,246 MW, the lowest among the eight evaluated configurations.

Figure 2 compares the ND MAE values across the candidate models. The LSTM achieved the second-lowest error at 1,531 MW, followed by CNN-GRU (1,563 MW) and Bi-RNN (1,576 MW). Higher errors were recorded for Bi-GRU (1,799 MW), CNN-LSTM (2,011 MW), Bi-LSTM (2,108 MW), and GRU (2,169 MW). The difference between the conventional RNN and the other configurations was therefore evident in the ND results, although the magnitude of that difference varied among architectures.

The same general pattern was observed for TSD and EWD. However, the relative performance of the non-RNN models differed between the two series. These findings indicate that the conventional RNN was the strongest performer under the implemented training and evaluation procedures, while additional gating, bidirectional processing, or convolutional feature extraction did not consistently produce lower prediction errors.

Table 3. Test-set forecasting errors of eight sequence-learning architectures for three electricity demand series

Model	ND			TSD			EWD		
	MSE	RMSE	MAE	MSE	RMSE	MAE	MSE	RMSE	MAE
RNN	2,145,599	1,465	1,151	3,604,549	1,899	1,517	1,816,562	1,348	1,069
Bi-RNN	3,965,580	1,991	1,576	4,534,598	2,129	1,747	2,622,502	1,619	1,312
LSTM	3,792,830	1,948	1,531	4,969,600	2,229	1,814	2,668,351	1,634	1,293
Bi-LSTM	6,234,142	2,497	2,108	5,483,910	2,342	1,901	7,014,201	2,648	2,309
GRU	7,756,762	2,785	2,169	9,667,411	3,109	2,444	5,967,167	2,443	1,943
Bi-GRU	4,862,814	2,205	1,799	6,220,262	2,494	2,082	3,485,477	1,867	1,550
CNN-LSTM	6,354,857	2,521	2,011	7,792,289	2,791	2,276	4,476,493	2,116	1,753
CNN-GRU	4,141,452	2,035	1,563	5,372,836	2,318	1,823	3,028,493	1,740	1,369

Note: MSE is expressed in MW^2, whereas RMSE and MAE are expressed in MW. Lower values indicate better forecasting performance; RNN = recurrent neural network; Bi- = bidirectional; LSTM = long short-term memory; GRU = gated recurrent unit; CNN = convolutional neural network; MSE = mean squared error; RMSE = root mean squared error; MAE = mean absolute error; ND = National Demand; TSD = Transmission System Demand; EWD = England and Wales Demand.

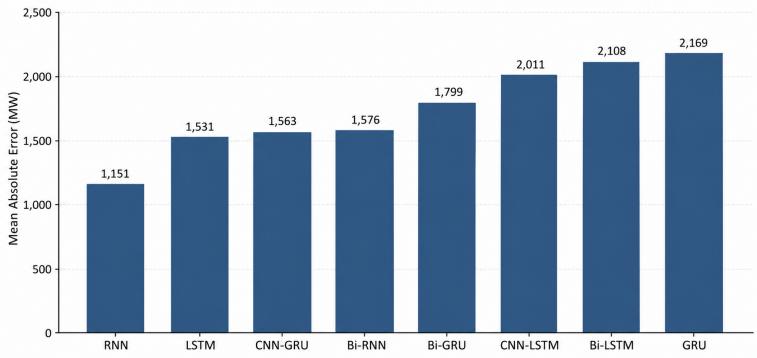


Figure 2. Comparison of test-set MAEs for ND across the eight sequence-learning architectures

Note: RNN = recurrent neural network; Bi- = bidirectional; LSTM = long short-term memory; GRU = gated recurrent unit; CNN = convolutional neural network; MAE = mean absolute error; ND = National Demand.

4.2 Performance Across Demand Series

The relative performance of the models was examined across the three demand targets to determine whether the observed results were confined to a particular series. The conventional RNN ranked first for ND, TSD, and EWD according to MAE, RMSE, and MSE. This consistency was notable because the three demand measures differed in their average magnitudes and geographical or operational coverage.

The rankings of the remaining architectures were less uniform. For example, LSTM produced a lower MAE than Bi-RNN for ND and EWD, whereas Bi-RNN performed better for TSD. CNN-GRU achieved relatively low errors across the three targets but did not consistently outperform the standalone LSTM or Bi-RNN configurations. The GRU model produced the highest MAE for ND and TSD, while Bi-LSTM recorded the highest error for EWD.

Figure 3 compares the observed and RNN-predicted demand values over the chronological test period. The predicted trajectories followed the main temporal variations in all three series, including the broad seasonal changes and periods of relatively high winter demand. Some differences between observed and predicted values remained, particularly around local fluctuations and demand peaks. The visual agreement was consistent with the comparatively low errors reported for the RNN in Table 3.

The results show that the conventional RNN performed consistently across the three related forecasting targets within this dataset. However, the similarity of the target series and their shared historical observations means that these results should not be interpreted as independent evidence of model performance across unrelated electricity systems.

4.3 Recursive Forecasting Over 366 Days

Following the test-set evaluation, the eight trained architectures were used to generate recursive demand forecasts for the period from 1 December 2023 to 30 November 2024. The forecasting procedure used a fixed 12-day input window, with each newly predicted demand vector incorporated into the input for the next forecasting step. Consequently, the models relied progressively on their own predictions rather than on newly observed demand values.

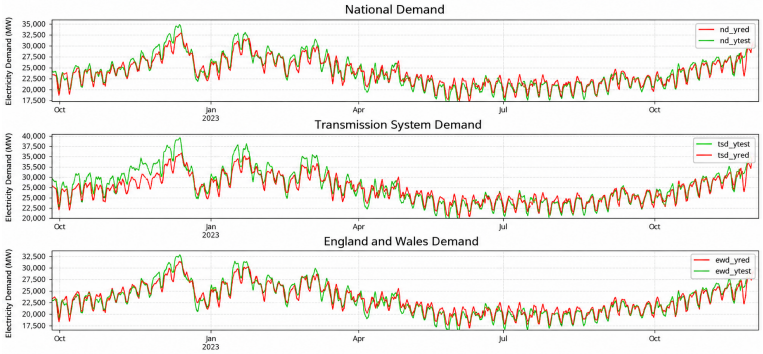


Figure 3. Observed and RNN-predicted ND, TSD, and EWD over the chronological test period

Note: RNN = recurrent neural network; ND = National Demand; TSD = Transmission System Demand; EWD = England and Wales Demand.

The resulting trajectories generally retained the broad seasonal structure present in the historical data, with higher predicted demand during winter and lower demand during summer. Differences were observed in the degree of variation maintained by individual architectures over the extended forecasting period. The recursive forecasting exercise was used to examine the behaviour of the models beyond the observed input period.

These observations describe the behaviour of the recursive forecasts rather than their independently measured accuracy. Because quantitative comparisons with observed demand were not available for the 366-day period, the projections could not establish which architecture provided the most accurate long-horizon forecasts. The recursive exercise instead illustrates how the forecasting configurations behaved when prediction errors and changes in the input sequence could propagate across successive forecasting steps.

5 Discussion

5.1 Interpretation of Model Performance

The conventional RNN achieved the lowest forecasting errors across the three electricity demand series, whereas the gated, bidirectional, and convolutional-recurrent configurations did not consistently improve prediction accuracy. This result is relevant to the selection of forecasting models for electricity systems, where the suitability of an architecture depends on the available observations, temporal resolution, and intended forecasting horizon. Several characteristics of the present experiment may help explain the observed differences.

First, the dataset contained 2,160 daily observations, a relatively limited number for training the deeper recurrent and hybrid networks included in the comparison. Models with larger numbers of trainable parameters may require more observations or stronger regularisation to estimate their parameters reliably. The RNN configuration may therefore have been better suited to the available dataset, although this explanation remains tentative because model capacity and overfitting were not examined under controlled conditions.

Second, daily aggregation removed much of the short-term variation present in the original half-hourly records. The resulting demand series retained broad seasonal patterns and temporal dependence but contained less intraday detail. Under these conditions, the additional representational capacity of gated or convolutional architectures may not have translated into lower forecasting errors. The findings therefore concern daily-aggregated electricity demand and should not be extended directly to hourly or sub-hourly forecasting tasks.

Third, the GRU and Bi-GRU configurations used temporal averaging to convert sequence outputs into next-day predictions, whereas the remaining architectures produced their final prediction vectors directly. This difference introduced an additional processing operation that may have affected the resulting errors. Its contribution cannot be isolated from the present results, and the relatively poor performance of these configurations should not be attributed solely to their recurrent cell structures.

Finally, the architectures were trained using different learning rates, maximum epoch counts, and network configurations. The deeper hybrid models received shorter maximum training durations than several recurrent alternatives. These differences may have affected convergence and generalisation, although the available results do not establish whether individual models were undertrained or overfitted. The reported ranking should therefore be understood as a comparison of the implemented configurations rather than a definitive assessment of architectural superiority.

Taken together, the findings indicate that increasing network complexity is not sufficient to guarantee lower forecasting errors. For daily electricity demand data of limited size, simpler recurrent configurations warrant consideration alongside more elaborate alternatives. Reliable model selection nevertheless requires evaluation procedures that account for training conditions, output construction, and the characteristics of the forecasting task.

5.2 Implications for Sustainable Power System Management

Electricity demand forecasts provide information for generation planning, system balancing, and the management of flexible electricity consumption. In power systems with increasing contributions from wind and solar generation, demand information can be considered alongside renewable generation forecasts, storage availability, and network constraints when assessing future operating conditions. The present findings are relevant to this process because they show that a comparatively simple recurrent model produced lower prediction errors than several more complex alternatives for three related demand measures in Great Britain.

The multi-output framework also has potential operational relevance. ND, TSD, and EWD describe related but distinct aspects of electricity system requirements. Their simultaneous prediction may support analyses that require consistent information across several demand measures. However, the present study did not compare multi-output forecasting with independently trained single-output models, and no direct improvement attributable to parameter sharing can therefore be established.

The findings may also inform subsequent work on demand-side management. Industrial electricity consumers, including computer numerical control machining and additive-manufacturing facilities, can exhibit time-dependent electricity requirements that interact with production schedules and operational constraints. Forecasting methods examined at the electricity system level may provide a starting point for investigating such applications, particularly where available training datasets are limited. Nevertheless, facility-level demand differs from national electricity demand in its temporal variability, operating conditions, and dependence on individual production processes. Direct transfer of the reported model ranking would therefore require separate validation using industrial electricity consumption records.

The study did not quantify renewable energy curtailment, electricity cost reductions, or carbon emissions. The study provides empirical evidence for selecting sequence-learning configurations in electricity demand forecasting. Assessing their effects on system operation, renewable energy utilisation, and emissions would require integration with dispatch or demand-response models.

5.3 Study Limitations

Several limitations should be considered when interpreting the results. The forecasting models were evaluated using a single chronological training–test partition, without rolling-origin validation or repeated training experiments. The reported errors therefore describe performance over one historical test period, and the stability of the model rankings across other periods remains unverified.

The models also relied exclusively on historical demand observations. External predictors, including temperature, weather conditions, calendar effects, electricity prices, and renewable generation, were not incorporated. Their omission limits the ability of the models to represent demand changes associated with external conditions and restricts the interpretation of their operational usefulness.

A further limitation concerns the 366-day recursive forecasting exercise. Because predicted values were repeatedly used as subsequent inputs, errors could propagate over the forecasting horizon. Quantitative accuracy against observed demand for that period was not reported, so the resulting trajectories should be regarded as exploratory long-horizon projections rather than validated operational forecasts.

Finally, the model comparison was affected by differences in architecture size, training configuration, and output handling. In particular, temporal averaging was applied only to the GRU and Bi-GRU predictions. These differences prevent the observed ranking from being attributed exclusively to the underlying neural network architectures. Future experiments should use consistent next-step output construction, comparable hyperparameter-selection procedures, and repeated chronological evaluations to establish whether the reported differences persist under more tightly controlled conditions.

6 Conclusions

This study compared eight deep sequence-learning architectures for the simultaneous prediction of ND, TSD, and EWD using daily electricity demand records from Great Britain. Under the implemented training and evaluation procedures, the conventional RNN achieved the lowest forecasting errors across all three targets. Its MAEs were 1,151 MW for ND, 1,517 MW for TSD, and 1,069 MW for EWD, corresponding to an average of approximately 1,246 MW. The LSTM, GRU, bidirectional, and convolutional-recurrent configurations did not produce lower errors under the tested conditions. These findings indicate that more elaborate network structures do not necessarily provide greater predictive accuracy when applied to relatively limited, daily-aggregated electricity demand data. However, differences in training configurations and output processing prevent the observed ranking from being attributed solely to architectural characteristics.

The results provide empirical evidence for selecting sequence-learning models in electricity demand forecasting, a task relevant to the planning and operation of sustainable power systems. The multi-output framework allowed three related demand measures to be predicted within a common modelling structure, while the comparison identified a

relatively simple recurrent configuration that performed consistently across the targets. Such forecasting approaches may inform subsequent studies of generation planning, demand-side management, and the coordination of electricity consumption with variable renewable generation. Their operational value, however, requires further assessment using additional system information and application-specific models. The present study did not directly evaluate electricity dispatch, renewable energy curtailment, energy cost reductions, or carbon emissions.

Further research should examine the stability of model rankings through rolling-origin validation and repeated training experiments, incorporate weather and calendar variables, and apply consistent output handling and comparable hyperparameter-selection procedures across architectures. Independent evaluation of the 366-day recursive forecasts against observed demand is also needed to establish their long-horizon accuracy. Probabilistic forecasting may provide additional information on prediction uncertainty for operational planning. Finally, extending the comparison to facility-level electricity consumption data would allow its applicability to industrial demand management and energy-aware production scheduling to be examined under realistic operating constraints.

Author Contributions

Conceptualization, A.A.; methodology, A.A., S.G., and S.J.; software, S.G. and S.J.; validation, A.A., S.G., and S.J.; formal analysis, A.A., S.G., and S.J.; investigation, A.A., S.G., and S.J.; data curation, S.G. and S.J.; writing—original draft preparation, A.A., S.G., and S.J.; writing—review and editing, A.A.; visualization, S.G. and S.J.; supervision, A.A.; project administration, A.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The electricity-demand data supporting the findings of this study are publicly available in the Kaggle repository "Electricity consumption UK 2009–2022," cited in Reference [4]. The model configurations and preprocessing procedures are described in the manuscript, and additional analysis materials are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

Generative AI tools, including Claude (Anthropic) and ChatGPT (OpenAI), were used solely to assist with language polishing, grammar checking, and manuscript formatting. The tools were not used to generate research data, conduct analyses, or develop scientific conclusions. All AI-assisted edits were reviewed and verified by the authors, who retain full responsibility for the content and accuracy of the manuscript.

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Nomenclature

ND	National Demand (MW)
TSD	Transmission System Demand (MW)
EWD	England and Wales Demand (MW)
RNN	Recurrent neural network
LSTM	Long short-term memory
GRU	Gated recurrent unit
CNN	Convolutional neural network
N	Number of windowed training examples

Error metrics

MSE	Mean squared error
RMSE	Root mean squared error
MAE	Mean absolute error
MAPE	Mean absolute percentage error

Symbols

x	Raw (unscaled) series value
x'	Min-Max-scaled series value
y_i,$\hat{y}_i$	True and predicted value for observation i
n	Number of test-set observations